

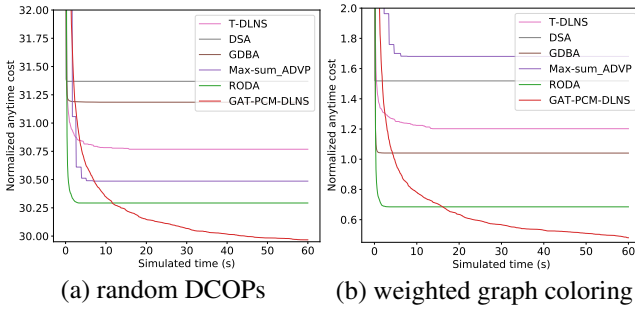


Figure 4: Convergence analysis

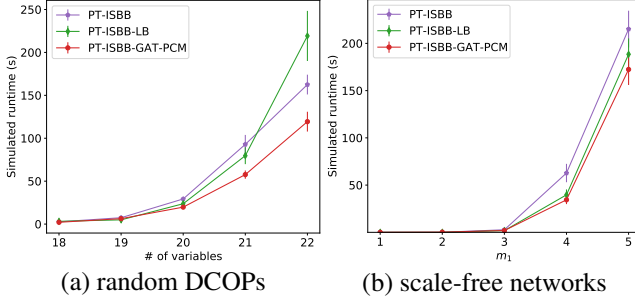


Figure 5: Runtime comparison on the problems with $d = 5$

to the number of agents and the size of each constraint matrix (cf. Prop. 2). It can be concluded that our method has a modest memory requirement and scales up to large instances well in various settings. In particular, our method has a (nearly) constant memory footprint when solving grid network problems since each agent is constrained with at most four other agents regardless of the grid size.

To investigate how fast our GAT-PCM-DLNS finds a good solution, we conduct a convergence analysis which measures the performance in terms of simulated time (?) on the problems with $|I| = 1000, p_1 = 0.005$ and $d = 3$ and present the results in Fig. 4. It can be seen that local search algorithms including DSA and GDBA quickly converge to a poor local optimum, while RODA finds a better solution in the first three seconds. T-DLNS slowly improves the solution but is strictly dominated by RODA. In contrast, our GAT-PCM-DLNS improves much steadily, outperforming all baselines after 18 seconds.

Finally, we demonstrate the merit of our GAT-PCM in accelerating backtracking search for random DCOPs with $p_1 = 0.25$ and scale-free networks with $|I| = 18, m_0 = 5$ by conducting a case study on the symmetric version of PT-ISABB (?) (referred as PT-ISBB) and present the results in Fig. 5. Specifically, we set the memory budget $k = 2$ and compare the simulated runtime of PT-ISBB using three domain ordering generation techniques: alphabetically, lower bound lookup tables (PT-ISBB-LB), and our GAT-PCM (PT-ISBB-GAT-PCM). For PT-ISBB-GAT-PCM, we only perform domain ordering for the variables in the first three levels in a pseudo tree. It can be observed that the backtracking search with alphabetic domain ordering performs poorly and is dominated by the one with the lower bound induced

domain ordering in the most cases. Notably, when solving the problems with 22 variables, PT-ISBB-LB exhibits the worst performance, because the lower bounds generated by approximated inference are not tight in complex problems, and hence the induced domain ordering may not prioritize promising assignments properly. On the other hand, our GAT-PCM powered backtracking search uses the predicted total cost of a subproblem as the criterion, resulting in a more efficient domain ordering and thus achieving the best results in solving complex problems.

Conclusion

In this paper, we present GAT-PCM, the first effective and general purpose deep pretrained model for DCOPs. We propose a novel directed acyclic graph representation schema for DCOPs and leverage the Graph Attention Networks (GATs) to embed our graph representations. Instead of generating heuristics for a particular algorithm, we train the model with optimally labelled data to predict the optimal cost of a target assignment given a partial assignment, such that GAT-PCM can be applied to boost the performance of a wide range of DCOP algorithms where evaluating the quality of an assignment is critical. To enable efficient graph embedding in a distributed environment, we propose DES to perform decentralized model inference without disclosing local constraints, where each agent exchanges only the embedded vectors via localized communication. Finally, we develop several heuristics based on GAT-PCM to improve local search and backtracking search algorithms. Extensive empirical evaluations confirm the superiority of GAT-PCM based algorithms over the state-of-the-arts.

In future, we plan to extend GAT-PCM to deal with the problems with higher-arity constraints and hard constraints. Besides, since agents in DES exchange the embedded vectors instead of constraint costs, it is promising to extend our methods to an asymmetric setting (?).

Acknowledgement

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